



7.3.2 Gauss-Seidel iteration ¶

7.3.2 Gauss Seidel iteration

fit width



A variation on the Jacobi iteration is the Gauss-Seidel iteration. It recognizes that since values at points are updated in some order, if a neighboring value has already been updated earlier in the current step, then you might as well use that updated value. For our example from [Subsection 7.1.1](#) this is captured by the algorithm

```
for k = 0, ..., convergence
  for i = 0, ..., N x N - 1
    v_i^(k+1) = (h^2 phi_i + v_{i-N}^(k+1) + v_{i-1}^(k+1) + v_{i+1}^(k) + v_{i+N}^(k))/4
  endfor
endfor
```

modified appropriately for points adjacent to the boundary. This algorithm exploits the fact that $v_{i-N}^{(k+1)}$ and $v_{i-1}^{(k+1)}$ have already been computed by the time $v_i^{(k+1)}$ is updated. Once again, the superscripts are there to emphasize the iteration during which a value is updated. In practice, the superscripts can be dropped because of the order in which the computation happens.

Homework 7.3.2.1. Modify the code for [Homework 7.1.1.1](#) (what you now know as the Jacobi iteration) to implement the Gauss-Seidel iteration.

▼ [Solution](#)

[Assignments/Week07/answers/Poisson_GS_iteration.m](#).

When you execute the script, in the COMMAND WINDOW enter "RETURN" to advance to the next iteration.

You may also want to observe the Jacobi and Gauss-Seidel iterations in action side-by-side in [Assignments/Week07/answers/Poisson_Jacobi_vs_GS.m](#).

Homework 7.3.2.2. Here we repeat [\(7.3.1\)](#) for Jacobi's iteration applied to the example in [Subsection 7.1.1](#):

$$\begin{pmatrix} 4 & & & & & & & & \\ & 4 & & & & & & & \\ & & 4 & & & & & & \\ & & & 4 & & & & & \\ & & & & 4 & & & & \\ & & & & & 4 & & & \\ & & & & & & 4 & & \\ & & & & & & & 4 & \\ & & & & & & & & \ddots \end{pmatrix} \begin{pmatrix} v_0^{(k+1)} \\ v_1^{(k+1)} \\ v_2^{(k+1)} \\ v_3^{(k+1)} \\ v_4^{(k+1)} \\ v_5^{(k+1)} \\ v_6^{(k+1)} \\ v_7^{(k+1)} \\ v_8^{(k+1)} \\ \vdots \end{pmatrix} \quad (7.3.2)$$

$$= \begin{pmatrix} 0 & 1 & & & & & & & \\ 1 & 0 & 1 & & & & & & \\ & 1 & 0 & 1 & & & & & \\ & & 1 & 0 & & & & & \\ 1 & & & 0 & 1 & & & 1 & \\ & 1 & & & 1 & 0 & 1 & & \ddots \\ & & 1 & & & 1 & 0 & & \\ & & & 1 & & & & 0 & \ddots \\ & & & & \ddots & & & \ddots & \ddots \end{pmatrix} \begin{pmatrix} v_0^{(k)} \\ v_1^{(k)} \\ v_2^{(k)} \\ v_3^{(k)} \\ v_4^{(k)} \\ v_5^{(k)} \\ v_6^{(k)} \\ v_7^{(k)} \\ v_8^{(k)} \\ \vdots \end{pmatrix} + \begin{pmatrix} h^2 \phi_0 \\ h^2 \phi_1 \\ h^2 \phi_2 \\ h^2 \phi_3 \\ h^2 \phi_4 \\ h^2 \phi_5 \\ h^2 \phi_6 \\ h^2 \phi_7 \\ h^2 \phi_8 \\ \vdots \end{pmatrix}.$$

Modify this to reflect the Gauss-Seidel iteration.

▼ [Solution](#)

$$\begin{pmatrix}
4 & & & & & & & & \\
-1 & 4 & & & & & & & \\
& -1 & 4 & & & & & & \\
& & -1 & 4 & & & & & \\
-1 & & & & 4 & & & & \\
& -1 & & & -1 & 4 & & & \\
& & -1 & & & -1 & 4 & & \\
& & & -1 & & & -1 & 4 & \\
& & & & & \ddots & \ddots & \ddots &
\end{pmatrix}
\begin{pmatrix}
v_0^{(k+1)} \\
v_1^{(k+1)} \\
v_2^{(k+1)} \\
v_3^{(k+1)} \\
v_4^{(k+1)} \\
v_5^{(k+1)} \\
v_6^{(k+1)} \\
v_7^{(k+1)} \\
v_8^{(k+1)} \\
\vdots
\end{pmatrix}
=
\begin{pmatrix}
0 & 1 & & & & & & & \\
& 0 & 1 & & & & & & \\
& & 0 & 1 & & & & & \\
& & & 0 & 1 & & & & \\
& & & & 0 & 1 & & & \\
& & & & & 0 & 1 & & \\
& & & & & & 0 & 1 & \\
& & & & & & & 0 & \\
& & & & & & & & 0 & \ddots \\
& & & & & & & & & \ddots
\end{pmatrix}
\begin{pmatrix}
v_0^{(k)} \\
v_1^{(k)} \\
v_2^{(k)} \\
v_3^{(k)} \\
v_4^{(k)} \\
v_5^{(k)} \\
v_6^{(k)} \\
v_7^{(k)} \\
v_8^{(k)} \\
\vdots
\end{pmatrix}
+
\begin{pmatrix}
h^2 \phi_0 \\
h^2 \phi_1 \\
h^2 \phi_2 \\
h^2 \phi_3 \\
h^2 \phi_4 \\
h^2 \phi_5 \\
h^2 \phi_6 \\
h^2 \phi_7 \\
h^2 \phi_8 \\
\vdots
\end{pmatrix}.$$

This homework suggests the following:

- We wish to solve $Ax = y$.

We write symmetric A as

$$A = \underbrace{(D - L)}_M - \underbrace{(L^T)}_N,$$

where $-L$ equals the strictly lower triangular part of A and D is its diagonal.

- We then notice that

$$Ax = y$$

can be rewritten as

$$(D - L - L^T)x = y$$

or, equivalently,

$$(D - L)x = L^T x + y.$$

If you think about it carefully, this captures (7.3.2) for our example. Finally,

$$x = (D - L)^{-1}(L^T x + y).$$

- If we now let $x^{(k)}$ be the values of our vector x in the current step. Then the values after all elements have been updated are given by the vector

$$x^{(k+1)} = (D - L)^{-1}(L^T x^{(k)} + y).$$

Homework 7.3.2.3. When the Gauss-Seidel iteration is used to solve $Ax = y$,

where $A \in \mathbb{R}^{n \times n}$, it computes entries of $x^{(k+1)}$ in the forward order

$x_0^{(k+1)}, x_1^{(k+1)}, \dots$. If $A = D - L - L^T$, this is captured by

$$(D - L)x^{(k+1)} = L^T x^{(k)} + y. \quad (7.3.3)$$

Modify (7.3.3) to yield a "reverse" Gauss-Seidel method that computes the

entries of vector $x^{(k+1)}$ in the order $x_{n-1}^{(k+1)}, x_{n-2}^{(k+1)}, \dots$.

▼ [Solution](#)

The reverse order is given by $x_{n-1}^{(k+1)}, x_{n-2}^{(k+1)}, \dots$. This corresponds to the splitting $M = D - L^T$ and $N = L$ so that

$$(D - L^T)x^{(k+1)} = Lx^{(k)} + y.$$

Homework 7.3.2.4. A "symmetric" Gauss-Seidel iteration to solve symmetric

$Ax = y$, where $A \in \mathbb{R}^{n \times n}$, alternates between computing entries in forward and reverse order. In other words, if $A = M_F - N_F$ for the forward Gauss-Seidel method and $A = M_R - N_R$ for the reverse Gauss-Seidel method, then

$$\begin{aligned}
M_F x^{(k+\frac{1}{2})} &= N_F x^{(k)} + y \\
M_R x^{(k+1)} &= N_R x^{(k+\frac{1}{2})} + y
\end{aligned}$$

constitutes one iteration of this symmetric Gauss-Seidel iteration. Determine M and N such that

$$Mx^{(k+1)} = Nx^{(k)} + y$$

equals one iteration of the symmetric Gauss-Seidel iteration.

(You may want to follow the hint...)

▼ [Hint](#) ▼ [Solution](#)

- From this unit and the last homework, we know that $M_F = (D - L)$, $N_F = L^T$, $M_R = (D - L^T)$, and $N_R = L$.

- Show that

$$(D - L^T)x^{(k+1)} = L(D - L)^{-1}L^Tx^{(k)} + (I + L(D - L)^{-1})y.$$

We show this by substituting M_R and N_R :

$$(D - L^T)x^{(k+1)} = Lx^{(k+\frac{1}{2})} + y$$

and then substituting in for $x^{(k+\frac{1}{2})}$, M_F and N_F :

$$(D - L^T)x^{(k+1)} = L((D - L)^{-1}(L^Tx^{(k)} + y)) + y.$$

Multiplying out the right-hand side and factoring out y yields the desired result.

- Show that $I + L(D - L)^{-1} = D(D - L)^{-1}$.

We show this by noting that

$$\begin{aligned} I + L(D - L)^{-1} &= \\ (D - L)(D - L)^{-1} + L(D - L)^{-1} &= \\ (D - L + L)(D - L)^{-1} &= \\ D(D - L)^{-1}. \end{aligned}$$

- Use these insights to determine M and N .

We now notice that

$$(D - L^T)x^{(k+1)} = L(D - L)^{-1}L^Tx^{(k)} + (I + L(D - L)^{-1})y$$

can be rewritten as (Someone check this... My brain hurts.)

$$\begin{aligned} (D - L^T)x^{(k+1)} &= L(D - L)^{-1}L^Tx^{(k)} + D(D - L)^{-1}y \\ \underbrace{(D - L)D^{-1}(D - L^T)}_M x^{(k+1)} &= \underbrace{(D - L)D^{-1}L(D - L)^{-1}L^T}_N x^{(k)} + y \end{aligned}$$

- From this unit and the last homework, we know that $M_F = (D - L)$, $N_F = L^T$, $M_R = (D - L^T)$, and $N_R = L$.

- Show that

$$(D - L^T)x^{(k+1)} = L(D - L)^{-1}L^Tx^{(k)} + (I + L(D - L)^{-1})y.$$

- Show that $I + L(D - L)^{-1} = D(D - L)^{-1}$.

- Use these insights to determine M and N .